

Qiniso Manqoba Mngomezulu

Software Engineering Graduate | Full Stack Developer | Business Intelligence

Johannesburg, South Africa | +27 67 552 3363 | manqobamngo20@outlook.com

LinkedIn: www.linkedin.com/in/qiniso-manqoba-mngomezulu-132b98377 |

GitHub: github.com/Mngomezuluza & github.com/QinisoMngo

PROFESSIONAL SUMMARY

Software engineering graduate and IT Intern with hands-on experience across full-stack development, business intelligence and enterprise platform support. Build and secure production features on a multi-tenant business platform (Next.js, TypeScript, PostgreSQL/Supabase, Vercel), and support a client KPI dashboard programme built on TIBCO Spotfire, TIBCO Data Virtualization (TDV) and an Enterprise Service Bus. Experienced in root cause analysis of server outages, business requirements documentation, and Microsoft 365, SharePoint, Active Directory and Entra ID administration. Strong foundation in C#, Java, JavaScript, Python and SQL from academic and personal projects across web, desktop, machine learning and VR.

TECHNICAL SKILLS

Languages: C#, Java, JavaScript, TypeScript, Python, SQL, PHP, PowerShell

Frameworks & Tools: .NET, ASP.NET, Blazor, React, Next.js, Node.js, Express, Prisma, NextAuth.js, Tailwind CSS, Git, GitHub, Postman, Vercel, Visual Studio, VS Code, Unity

Databases & APIs: SQL Server, PostgreSQL, Supabase (Storage, Row Level Security), MySQL, Oracle, MongoDB, Firebase, REST APIs, JSON, XML, OAuth, JWT, GraphQL

Business Intelligence & Data: TIBCO Spotfire, TIBCO Data Virtualization (TDV), KPI Dashboard Development, Business Requirements Documents (BRDs), Database Administration, Data Warehousing, Reporting

Integration: Enterprise Service Bus (TIBCO BusinessWorks), Boomi, Model Context Protocol (MCP) connectors

Servers & Incident Management: Windows Server (Spotfire, TDV and ESB servers), Windows Services, Tomcat configuration, JDK runtimes, TLS certificates & keystores, server log analysis, Root Cause Analysis (RCA)

Identity & Cloud Administration: Active Directory, Microsoft Entra ID, Microsoft 365 Admin Center, SharePoint Administration, Document360 (KMS), OneDrive-to-SharePoint Migration

Security & DevOps: Role-based access control, multi-tenant data scoping, audit logging, MFA, secret management, regression and integration testing, Vercel deployments and cron jobs, database backup and restore

AI-Assisted Development: Claude Code, GitHub Copilot, OpenAI Codex

SELECTED TECHNICAL PROJECTS

• PWD Group Portal – Multi-Tenant Business Management Platform (PWD Xperts)

Contributed to a production, multi-tenant platform spanning CRM, HR, finance, procurement, legal and tender management across the group's companies. Designed and built the role-based access control layer across HR, finance, hotel operations and tender modules; secured previously unprotected API routes with authentication, company scoping and audit logging, backed by regression tests. Improved dashboard performance through database indexing and query reduction, moved document uploads to Supabase Storage, reconciled conflicting finance calculations, and built spreadsheet and Sage data imports and an AI (MCP) connector for searching the tender library.

- **Tech stack:** TypeScript, Next.js, React, PostgreSQL, Prisma, Supabase, NextAuth.js, Tailwind CSS, Vercel, MCP

• Research Portal – Automated Survey & Reporting Platform (PWD Xperts)

Built an end-to-end automated bi-weekly reporting system for a PhD research survey platform: a cron-triggered pipeline that assembles a multi-document report pack from live survey data, with an admin review-and-approval workflow and email delivery. Also diagnosed and fixed a production-only chart-rendering bug by implementing custom vector-glyph label rendering with a bundled font.

- **Tech stack:** Next.js, React, TypeScript, Supabase, Recharts, docx, Vercel Cron

• KPI Dashboard Programme – Business Intelligence (Client Project, PWD Xperts)

Support the build and deployment of executive KPI dashboards for a public-sector client on TIBCO Spotfire, fed by TIBCO Data Virtualization and an Enterprise Service Bus. Write and revise Business Requirements Documents (current and target state, functional requirements, data sources, dashboard ownership) through review cycles with senior analysts, and support UAT and progress reporting.

- **Tech stack:** TIBCO Spotfire, TIBCO Data Virtualization (TDV), TIBCO BusinessWorks (ESB), Boomi, Windows Server

• VR Interactive Modelling Application

Built a Unity-based VR modelling application that let users draw and manipulate geometry with VR controllers, with snapping, undo/redo, and DXF export.

- **Tech stack:** Unity, C#, OpenXR, XR Interaction Toolkit

• LPR381 Linear & Integer Programming Solver

Integrated a C# Avalonia desktop interface with an F# algorithm engine to solve optimisation problems using methods such as Simplex, Branch and Bound, and Cutting Plane.

- **Tech stack:** C#, F#, Avalonia, .NET

• Maintenance Management App

Developed a full-stack maintenance tracking application with REST API workflows for creating, updating, filtering, and archiving service jobs.

Tech stack: Node.js, Express, MongoDB, JavaScript

• Poised Project Management System

Created a Java application for capturing, updating, searching, and finalising engineering project records using JDBC and SQL-driven CRUD operations.

- **Tech stack:** Java, JDBC, MariaDB, SQL

EXPERIENCE

IT Intern | PWD Xperts | July 2026 – Present

- Support Business Intelligence dashboard development for a client KPI dashboard programme, shadowing the Head of Dashboard Development; started TIBCO Spotfire and Spotfire Administration training in July 2026 and am learning TIBCO Data Virtualization (TDV) hands-on, alongside database administration and data warehousing tasks.
- Write and revise Business Requirements Documents (BRDs) for KPI dashboards, covering current and target state, functional requirements, data sources and dashboard ownership, and update them through review rounds with senior analysts.
- Contributed to a consolidated Root Cause Analysis (RCA) of production outages across Spotfire Server, TDV, the ESB and a web portal: gathered evidence from server logs, configuration files and timestamps, helped identify causes such as a disabled Tomcat class-path setting, a missing JDK runtime and a TLS certificate/keystore mismatch, and drafted remediation and monitoring recommendations.
- Take part in live troubleshooting sessions with infrastructure teams and vendors on the Windows servers hosting Spotfire, TDV and the ESB, and manage Active Directory and Microsoft Entra ID accounts and groups for BI platform access.
- Full-stack development on the company's in-house dashboard (full admin rights): designed and built the role-based access control layer across HR, finance, hotel operations and tender modules, and added authentication, company scoping and audit logging to previously unauthenticated API routes, backed by regression tests written to verify coverage.
- Diagnosed and resolved production issues on the in-house dashboard, including silent document-upload failures, a blocked database migration path, slow pages fixed through database indexing and fewer queries, and conflicting finance totals across modules.
- Designed and built an automated bi-weekly reporting system for the CEO's PhD research platform, from a cron-triggered data pipeline through to an admin review-and-approval workflow and email delivery; deployed the platform and report to the CEO on a bi-weekly basis.
- Took over ownership (full admin rights) of internal client projects following the departure of the Project Manager: SOC/Cybersecurity, Enterprise Service Bus (ESB), a client Customer Portal website, and the Knowledge Management System (Document360); update project plans, coordinate with the managed detection and response (MDR) provider, and log and follow up Boomi platform support tickets.
- Administer the Microsoft 365 Admin Center (full admin rights) and led the migration of company data from OneDrive to SharePoint, acting as SharePoint administrator.
- Shadow project leads/managers on documentation, distributing reports, and presenting monthly and weekly project reports; turn meeting outcomes into work-plan updates and risk registers, and collaborate with Tender Managers to integrate Sage Accounting data into the internal PWD Xperts dashboard.

Office Administrator & IT Support | SOWISO Consultants PTY LTD & SOWISO Foundation NPC | 2017-2020

- Provided technical support including software installation, troubleshooting, hardware setup, and day-to-day systems assistance.
- Maintained organizational records, supported administration processes, and used digital tools such as Outlook, Zoom, and Microsoft Teams.
- Automated repetitive work using Excel macros and basic scripting, while assisting with backups, IT inventory, and bookkeeping support.

EDUCATION

HyperionDev / Stellenbosch University | Feb 2026 – In Progress

Immersive Full Stack Web & Software Engineering Bootcamp

Degree - Belgium Campus iTversity | Jan 2023 – Dec 2025

Bachelor of Computing (Software Engineering)

Honours - Belgium Campus iTversity | Jan 2026 – In Progress

Bachelor of Computing (Software Engineering)

Eduvisa Graduate School of Management | Jan 2022 – Dec 2022

Graduate Certificate

St Benedict's College for Boys | Jan 2014 – Dec 18

National Senior Certificate | Core subjects: Mathematics, Information Technology, Business Studies, Accounting, English, Afrikaans

ADDITIONAL INFORMATION

Languages: English, Afrikaans, isiZulu, Setswana, isiXhosa

Achievement: Completed the Beyond Adventure gap-year programme (2022), reflecting discipline, resilience, leadership, and personal development.